

How should we form discussion groups?

Week 02, second hour

Business Agent Design

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What the questionnaire gave us

Name		Preferred group size	Named classmates
1	정현창	4	
2	김문건	3	
3	최현호		
4	이승욱		
...			
43	이승현	4	박주원, 장예지, 조수아
44	장예지	4	조수아, 이승현, 박주원
45	홍은기	3	신운호, 이원영

The full table is in `questionnaire.csv`

Form the discussion groups

5 min

Use the questionnaire table. Do not use AI.

- 1 Each group must contain 2–4 students
- 2 Every student must appear exactly once
- 3 Explain your decisions

Which group was easy?
Which was hard?

What do we have? What do we not know?

We have

Who is enrolled: 45 students

Who named whom: 56 requests,
from 31 students

Preferred group sizes: 38 answers

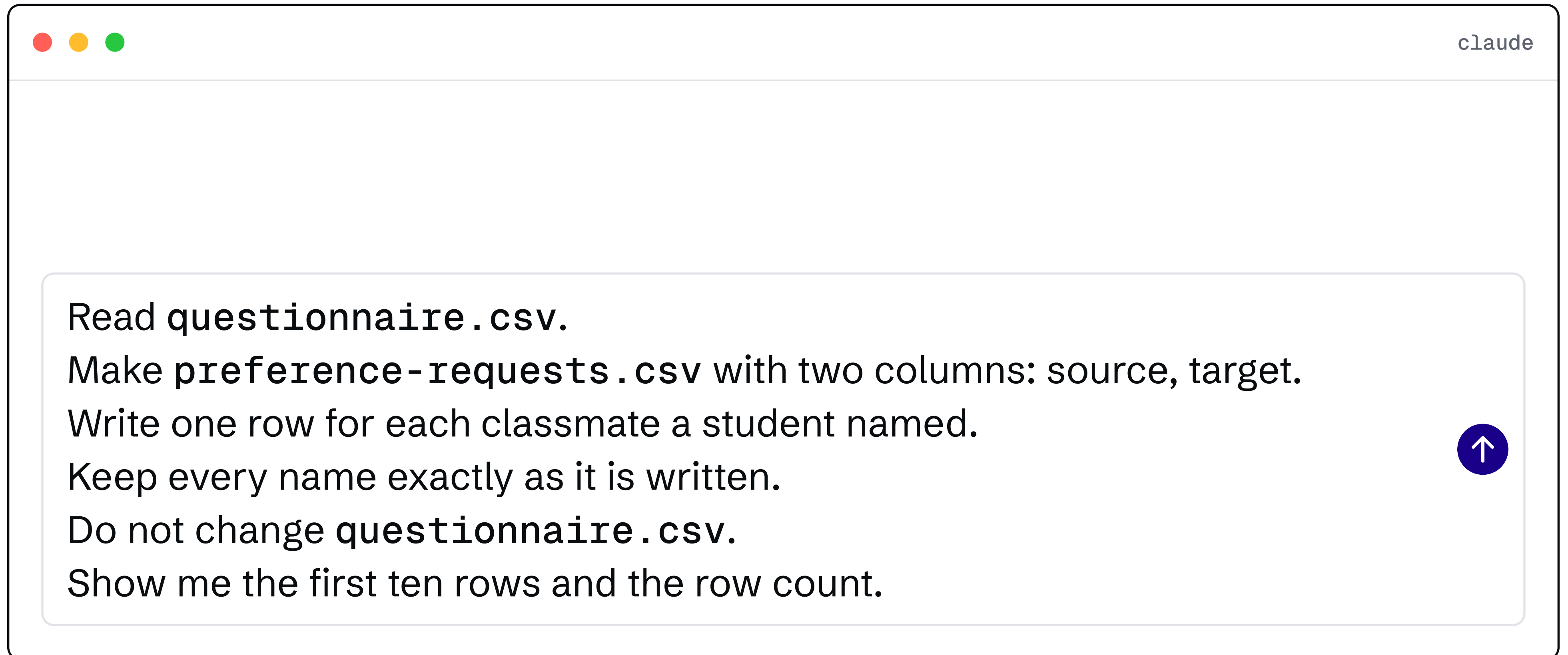
We do not know

What 14 students want: 7 did not
respond, 7 named nobody

How strongly each request is meant

Whether an unnamed pair would
work well together

Prompt 1: make the request list



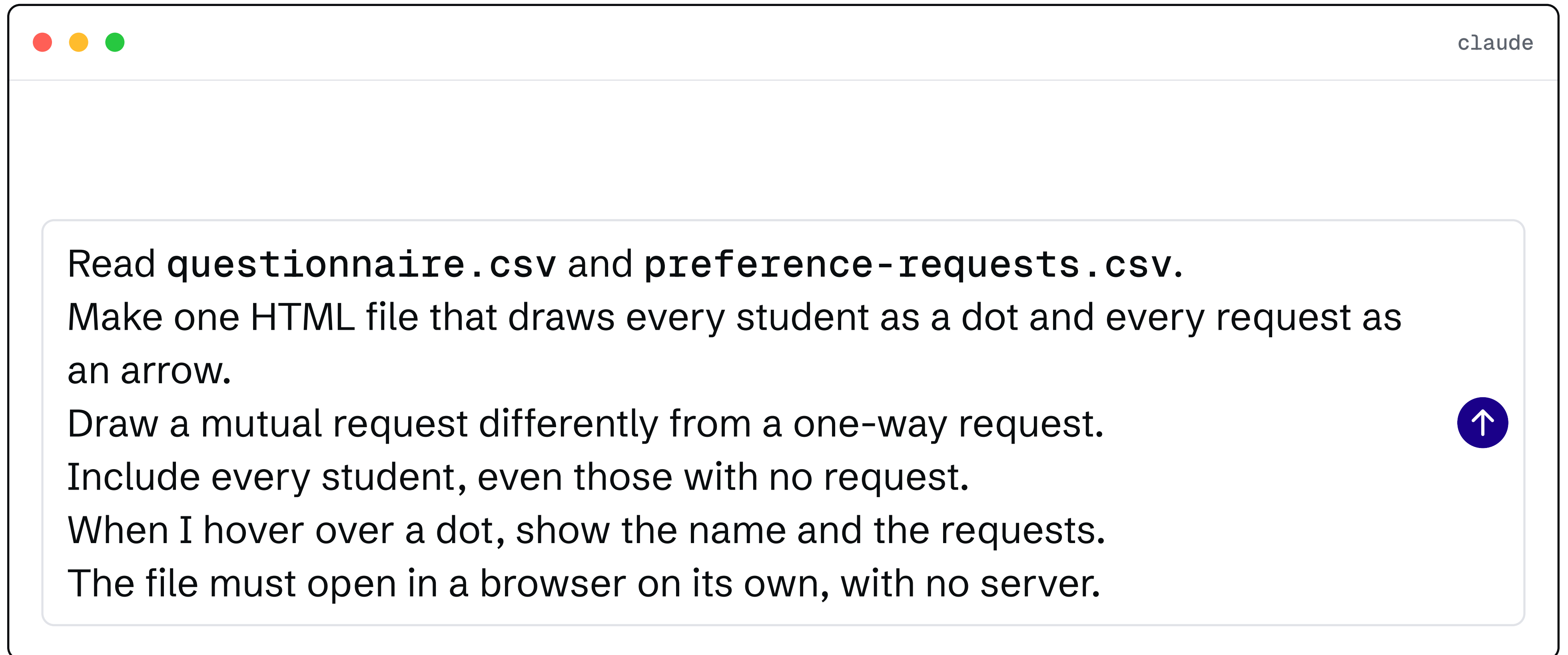
Every request becomes a row.



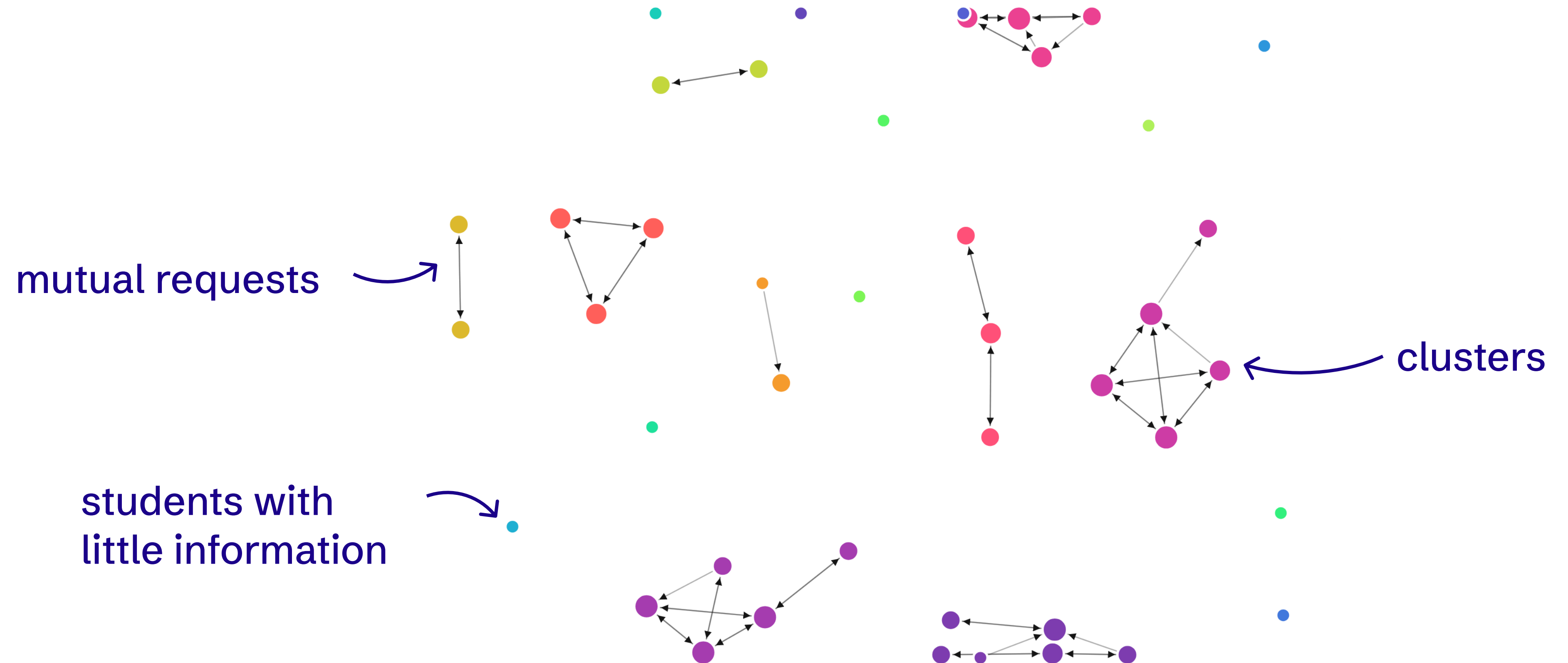
Name	Size	Named classmates
이승현	4	박주원, 장예지, 조수아

Source	Target
이승현	박주원
이승현	장예지
이승현	조수아

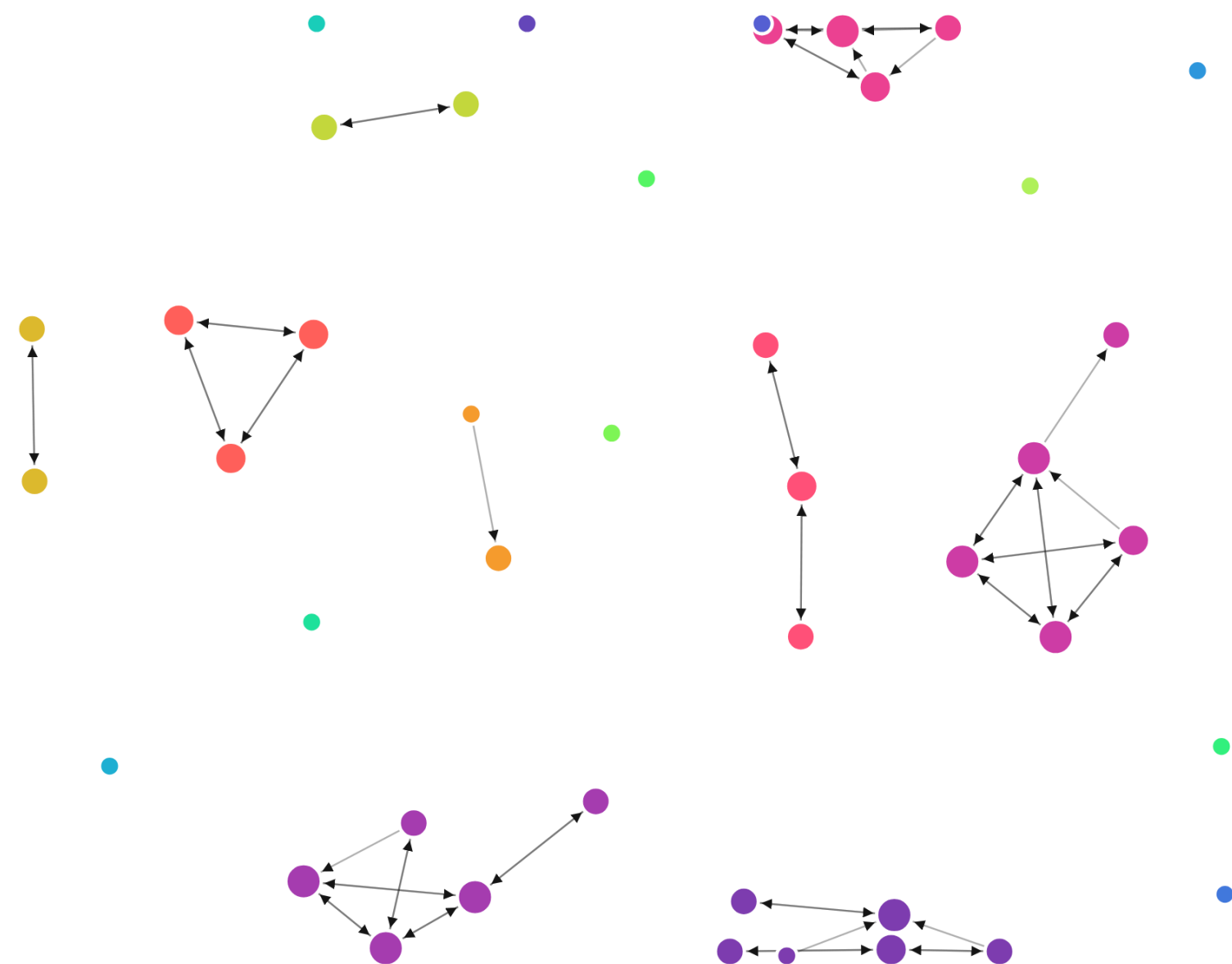
Prompt 2: draw the requests



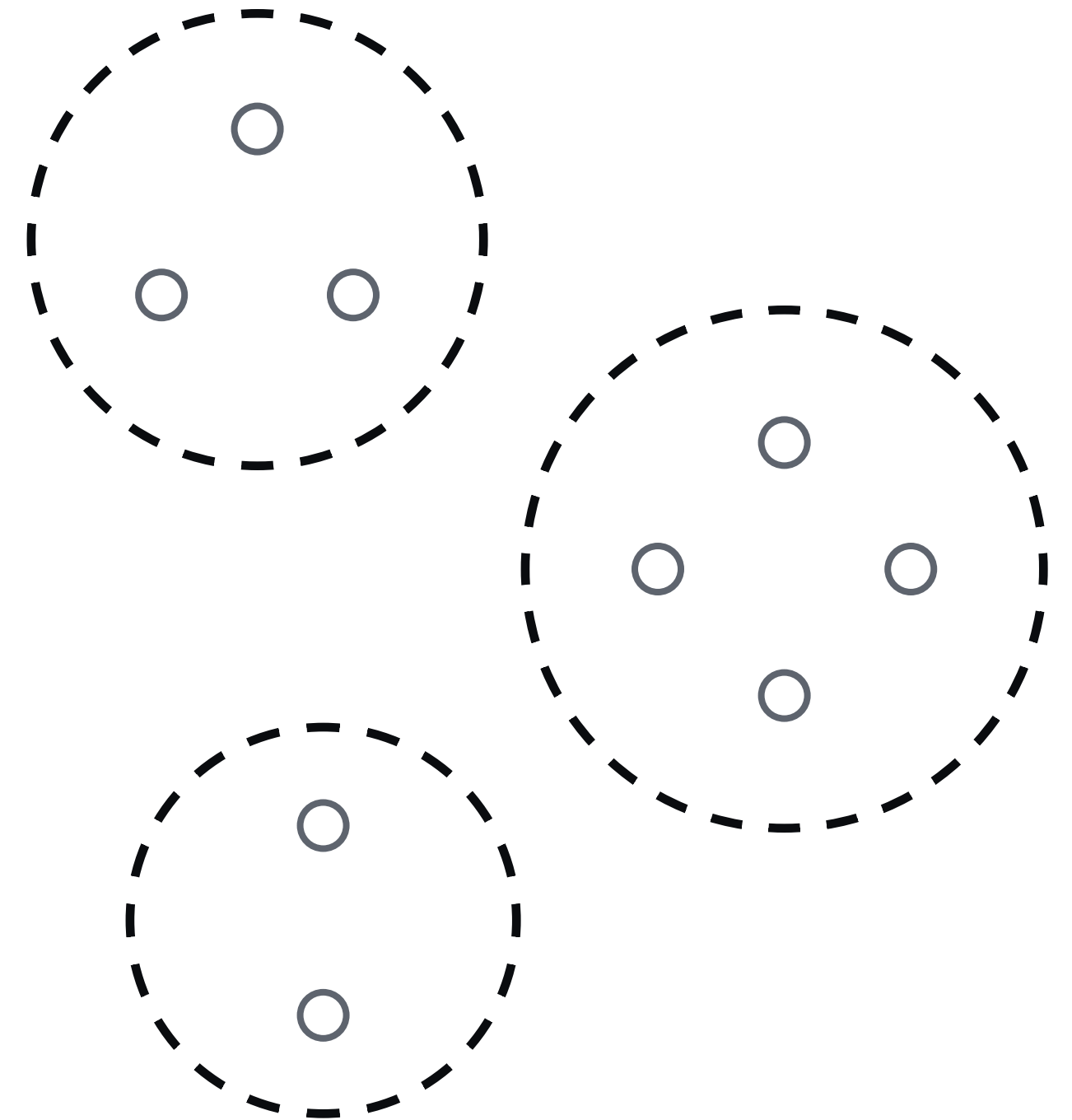
What do you notice?



The graph is not yet a grouping.



Who goes with whom?

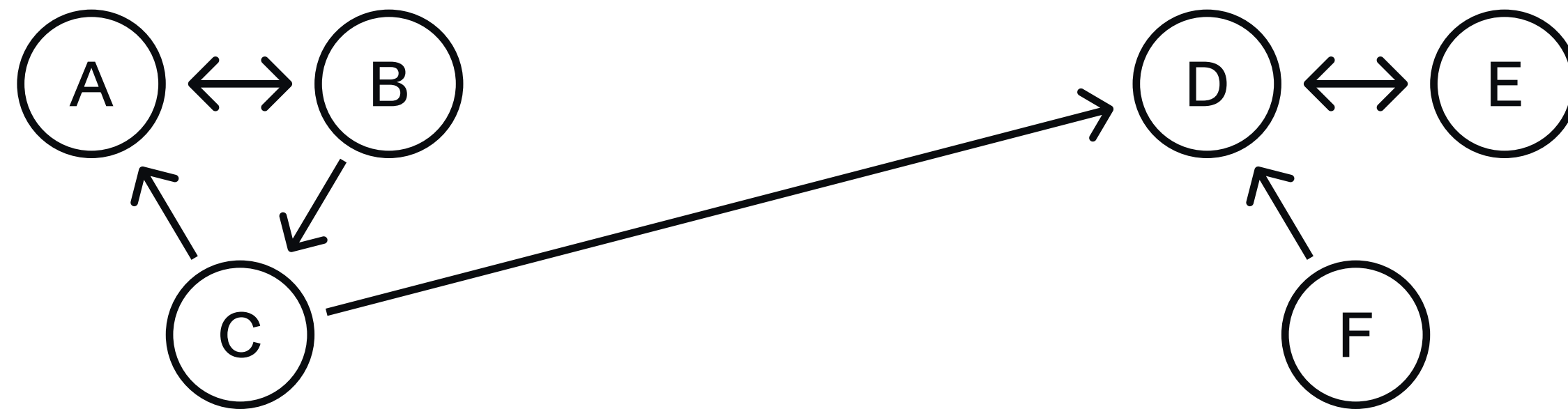


Constraints and objective, in words

Put every student in one group of 2 to 4.

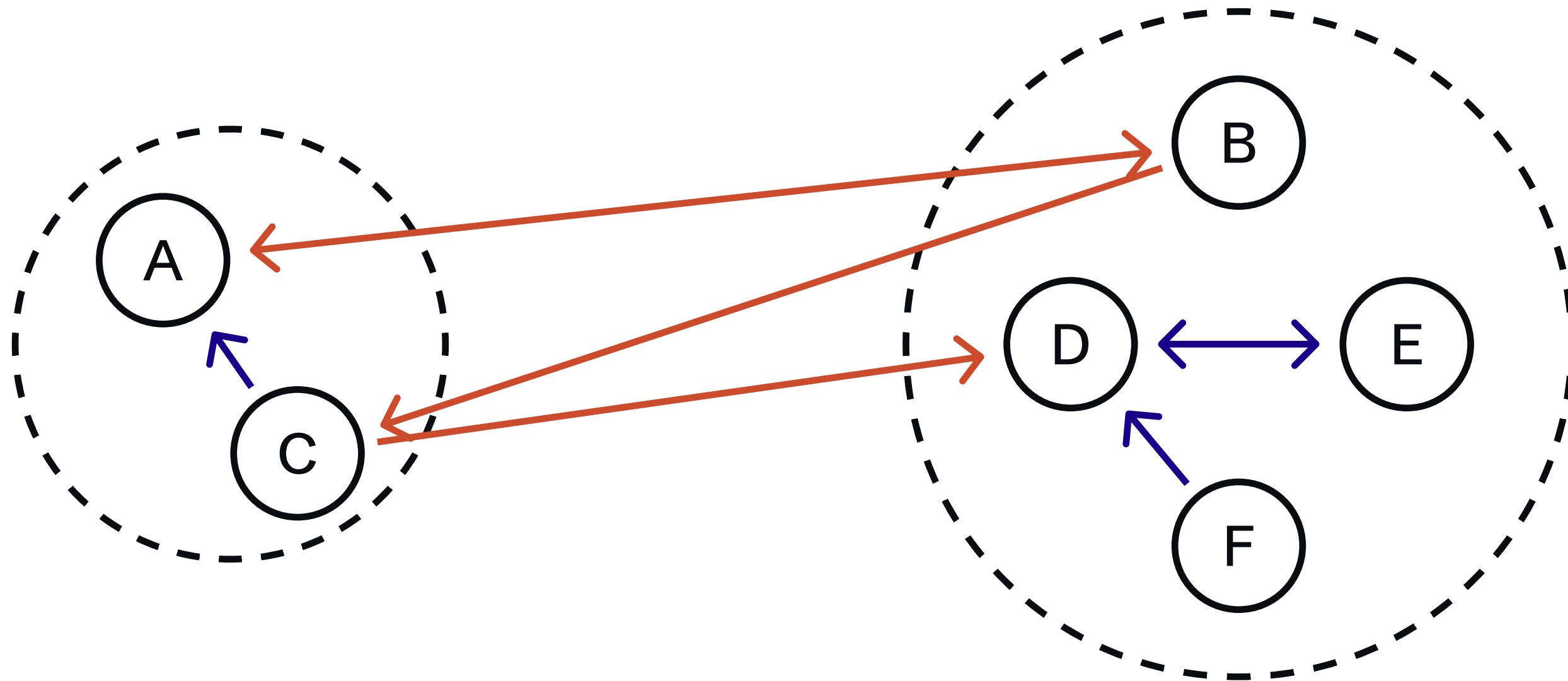
Keep as many named classmates together as possible.

A small example



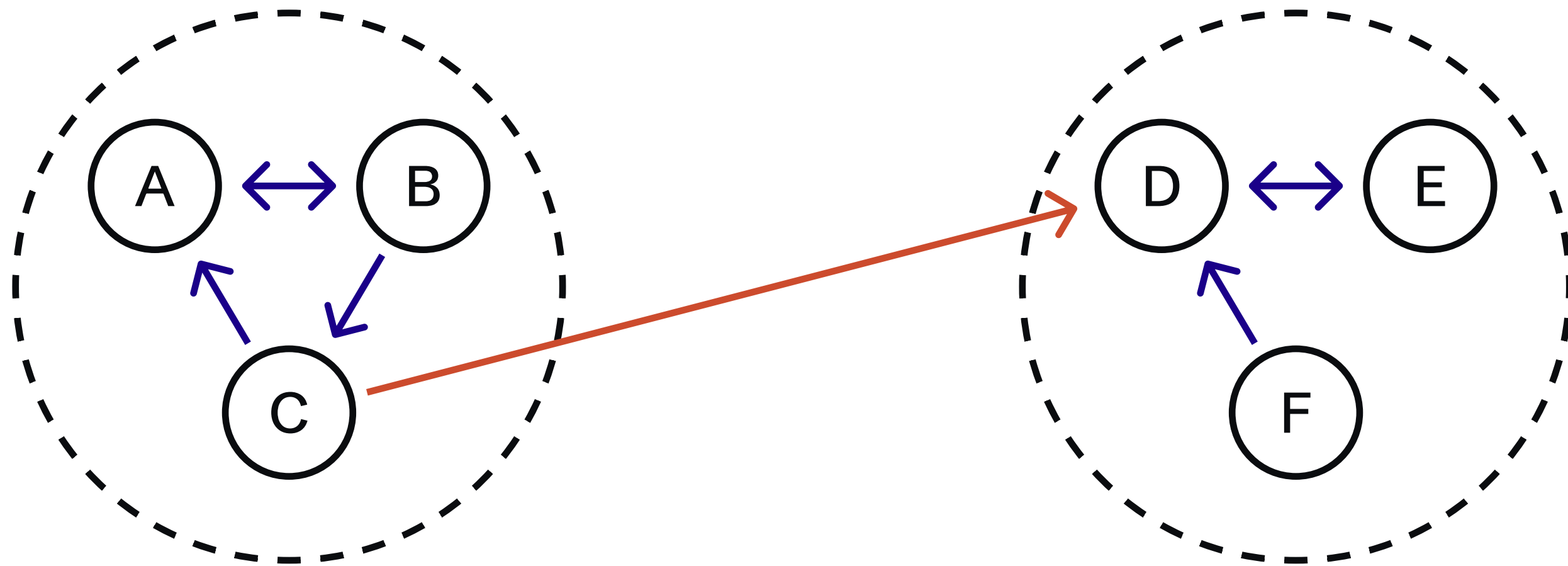
6 students · 8 requests · 2 groups of 2 to 4

One grouping



4 kept • 4 broken

A better grouping



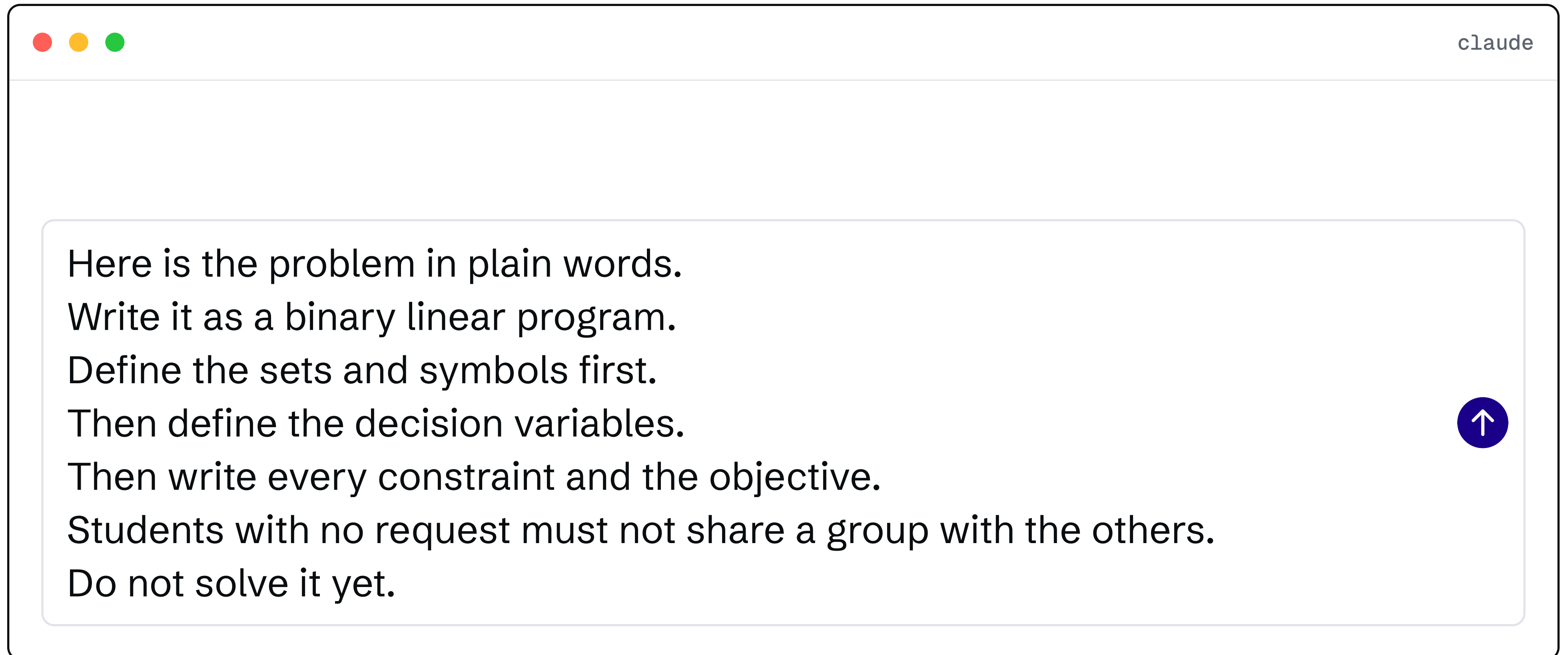
7 kept • 1 broken

**Is 7 the most we can keep?
How would we know?**

The group partitioning problem

- Take the students and their requests.
- Split them into K groups of 2 to 4.
- Keep as many requests inside a group as you can.
- Put students with no request at all in groups of their own (so every other group stays connected).

Prompt 3: write it as a program



The problem, formally

Given a set of students S , directed requests $R \subseteq S \times S$,
isolated students $I \subseteq S$ with $C = S \setminus I$, and K, L, U .

Find a partition of S into groups $G_1, \dots, G_K$

such that $L \leq |G_g| \leq U$ $\forall g$
 $G_g \subseteq I$ or $G_g \subseteq C$ $\forall g$

maximizing $|\{(i, j) \in R : i \text{ and } j \text{ share a group}\}|$

Decision variables

$$x_{ig} = \begin{cases} 1 & \text{if student } i \text{ is in group } g \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ijg} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are both in group } g \\ 0 & \text{otherwise} \end{cases}$$

$$z_g = \begin{cases} 1 & \text{if group } g \text{ is reserved for isolated students} \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_g x_{ig} = 1$$

every row

$$2 \leq \sum_i x_{ig} \leq 4$$

every column

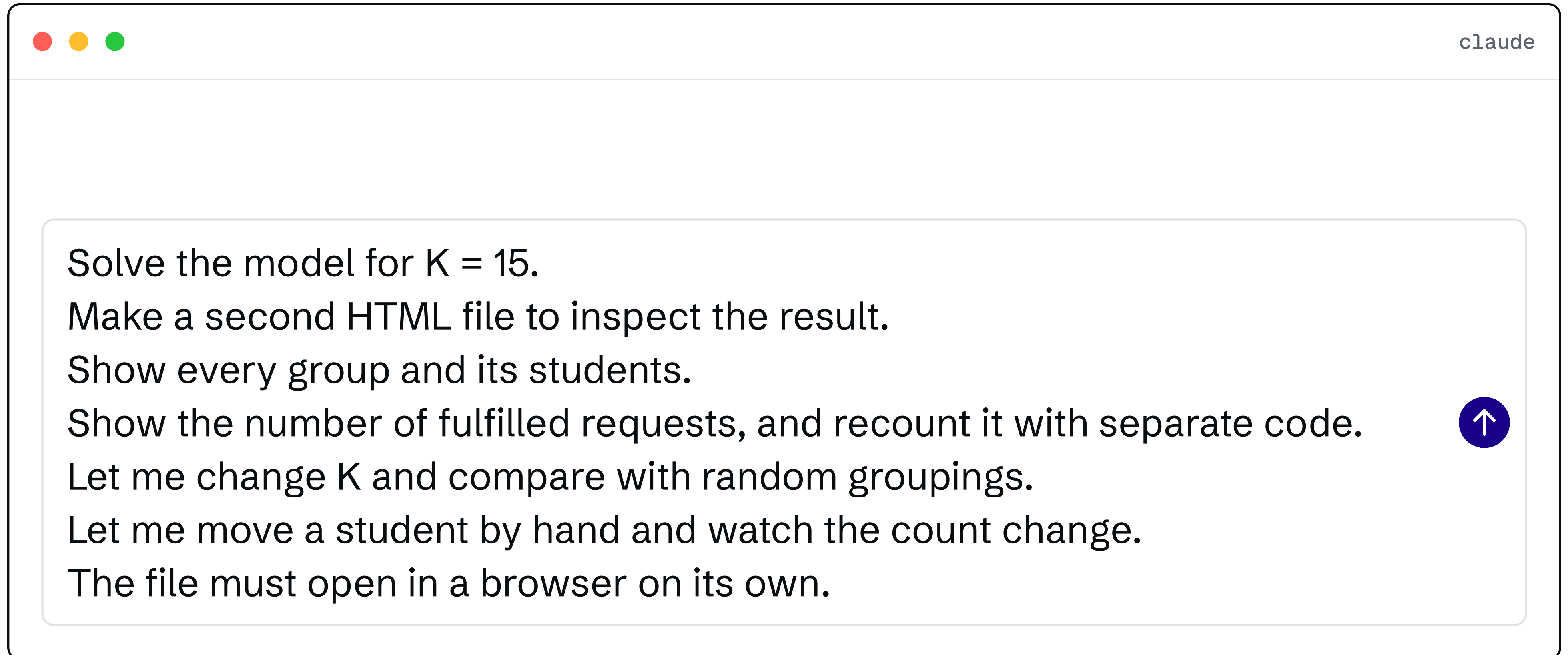
student	g = 1	g = 2	row sum
A	1	0	1
B	1	0	1
C	1	0	1
D	0	1	1
E	0	1	1
F	0	1	1
column sum	3	3	

The model

objective	max	$\sum_{(i,j) \in R} \sum_{g \in G_K} y_{ijg}$	
one group each	subject to	$\sum_{g \in G_K} x_{ig} = 1$	$\forall i \in S$
group size: $L = 2, U = 4$		$L \leq \sum_{i \in S} x_{ig} \leq U$	$\forall g \in G_K$
keep types apart		$x_{ig} \leq z_g$	$\forall i \in I, g \in G_K$
		$x_{ig} \leq 1 - z_g$	$\forall i \in C, g \in G_K$
$y = \text{kept request}$		$y_{ijg} \leq x_{ig}, \quad y_{ijg} \leq x_{jg}$	$\forall (i, j) \in R, g \in G_K$
		$y_{ijg} \geq x_{ig} + x_{jg} - 1$	$\forall (i, j) \in R, g \in G_K$
binary		$x_{ig}, y_{ijg}, z_g \in \{0, 1\}$	

How it is solved: a mixed-integer solver runs branch and bound. It reports the objective and a best bound. When the two are equal, the score is proven optimal.

Prompt 4: solve it and prove the score



Now we inspect the proposal

The HTML the agent built, on the live screen

What did our objective leave out?

Optimized

Fulfilled requests

Measured, not optimized

Student coverage

Preferred group size

Not measurable

Fairness

Unspoken reasons

The agent helped build a proposal. We still own the decision.

Human

define

inspect

decide

Agent

implement

test

explain

Next week we build the environment an agent works in, and learn to undo its changes.